1 Cauchy's Mean Value Theorem

To prove L'Hôpital's Rule, we will need the following generalization of the Mean Value Theorem.

Theorem 1.1 (Cauchy's Mean Value Theorem)**.** *Let f and g be functions such that*

(i) *f and g are continuous on $[a,b]$,*

(ii) *f and g are differentiable on (a,b), and*

(iii) *$g'(x) \neq 0$ for all $x \in (a,b)$.*

Then there exists $c \in (a,b)$ such that

$$\frac{f'(c)}{g'(c)} = \frac{f(b)-f(a)}{g(b)-g(a)}.$$

1. Show that in the special case in which $g(x) = x$, Theorem 1.1 is just the ordinary Mean Value Theorem.
   If $g(x) = x$, then the conclusion becomes that there exists $c \in (a,b)$ such that
   $$\frac{f'(c)}{1} = \frac{f(b)-f(a)}{b-a}.$$

2. Prove Theorem 1.1 by applying Rolle's Theorem to the function
   $$h(x) = f(x) - f(a) - \frac{f(b)-f(a)}{g(b)-g(a)}(g(x)-g(a)).$$
   Since f is differentiable on $[a,b]$ and continuous on (a,b), so is h. We also have
   $$\begin{aligned} h(b)-h(a) &= f(b)-f(a) - \frac{f(b)-f(a)}{g(b)-g(a)}(g(b)-g(a)) - (f(a)-f(a) - \frac{f(b)-f(a)}{g(b)-g(a)}(g(a)-g(a))) \\ &= f(b)-f(a) - \frac{f(b)-f(a)}{g(b)-g(a)}(g(b)-g(a)) - 0 \\ &= f(b)-f(a) - \frac{f(b)-f(a)}{g(b)-g(a)}(g(b)-g(a)) \\ &= f(b)-f(a)-(f(b)-f(a)) \\ &= 0. \end{aligned}$$
   Thus, h satisfies the hypotheses of Rolle's Theorem, so there exists $c \in (a,b)$ such that
   $$0 = h'(c) = f'(c) - \frac{f(b)-f(a)}{g(b)-g(a)}g'(c)$$
   and therefore
   $$\frac{f'(c)}{g'(c)} = \frac{f(b)-f(a)}{g(b)-g(a)}.$$

2 L'Hôpital's Rule

Theorem 2.1 (L'Hôpital's Rule)**.** *Suppose f and g are differentiable and $g'(x) \neq 0$ on an open interval I that contains a (except possibly at a). Suppose that*

$$\lim_{x\to a} f(x) = 0 \quad \textit{and} \quad \lim_{x\to a} g(x) = 0$$

or that

$$\lim_{x\to a} f(x) = \pm\infty \quad \textit{and} \quad \lim_{x\to a} g(x) = \pm\infty.$$

(In other words, we have an indeterminate form of type $\frac{0}{0}$ or $\frac{\infty}{\infty}$. Then

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}$$

if the limit on the right side exists (or is $\pm\infty$).

Proof. We will first prove L'Hôpital's Rule for indeterminate limits of type $\frac{0}{0}$. [1] Assume that $\lim_{x\to a} f(x) = \lim_{x\to a} g(x) = 0$ and let $L = \lim_{x\to a} \frac{f'(x)}{g'(x)}$. We need to show that $\lim_{x\to a} \frac{f(x)}{g(x)} = L$.

3. Define

$$F(x) = \begin{cases} f(x) & \text{if } x \neq a, \\ 0 & \text{if } x = a, \end{cases} \quad \text{and} \quad G(x) = \begin{cases} g(x) & \text{if } x \neq a, \\ 0 & \text{if } x = a. \end{cases}$$

   Show that F and G are both continuous on I.
   f, g are continuous on I by assumption, so F, G are continuous at each $x \neq a$. Since $\lim_{x\to a} F(x) = \lim_{x\to a} f(x) = 0 = F(a)$ and similarly for G, they are also continuous at $x = a$.

4. Let $x \in I$ and $x > a$. Show that the hypotheses of Cauchy's Mean Value Theorem hold for F and G on $[a, x]$.

   - Since F, G are continuous on I, in particular they are continuous on $[a, x]$.
   - Since f, g are differentiable on I, $F'(x) = f'(x)$ for $x \neq a$, and $G'(x) = g'(x)$ for $x \neq a$, F, G are differentiable on (a, x).
   - $G'(x) = g'(x) \neq 0$ by assumption on (a, x).

   This shows that the hypotheses of Cauchy's Mean Value Theorem hold for F, G on $[a, x]$.

[1] The proof of the other cases can be found at https://web.ma.utexas.edu/users/m408n/CurrentWeb/LM4-4-11.php.

5. Show that there is a number y such that $a < y < x$ and

$$\frac{F'(y)}{G'(y)} = \frac{F(x)}{G(x)}.$$

By Cauchy's Mean Value Theorem, there exists $y \in (a,x)$ such that

$$\frac{F'(y)}{G'(y)} = \frac{F(x)-F(a)}{G(x)-G(a)} = \frac{F(x)-0}{G(x)-0} = \frac{F(x)}{G(x)}.$$

6. Noting that $x \to a^+$ implies $y \to a^+$ (since $a < y < x$), use your answer above to show that

$$\lim_{x\to a^+} \frac{f(x)}{g(x)} = L.$$

$$\begin{aligned}
\lim_{x\to a^+} \frac{f(x)}{g(x)} &= \lim_{x\to a^+} \frac{F(x)}{G(x)} && \text{(since these agree when } x \neq a\text{)} \\
&= \lim_{x\to a^+} \frac{F'(y)}{G'(y)} && \text{(by part 5 above)} \\
&= \lim_{y\to a^+} \frac{F'(y)}{G'(y)} && \text{(since } x \to a^+ \text{ implies } y \to a^+\text{)} \\
&= \lim_{y\to a^+} \frac{f'(y)}{g'(y)} && \text{(since these agree when } x \neq a\text{)} \\
&= L.
\end{aligned}$$

A similar argument shows that $\lim_{x\to a^-} \frac{f(x)}{g(x)} = L$. Therefore $\lim_{x\to a} \frac{f(x)}{g(x)} = L$, proving L'Hôpital's Rule for the case where a is finite.

7. If a is infinite, let $t = \frac{1}{x}$. Then $t \to 0^+$ as $x \to \infty$, so

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = \lim_{t\to 0^+} \frac{f(1/t)}{g(1/t)}.$$

Use L'Hôpital's Rule for finite a to prove that this is equal to $\lim_{x\to\infty} \frac{f'(x)}{g'(x)}$, completing the proof for indeterminate limits of type $\frac{0}{0}$.

$$\begin{aligned}
\lim_{x\to\infty} \frac{f(x)}{g(x)} &= \lim_{t\to 0^+} \frac{f(1/t)}{g(1/t)} \\
&= \lim_{t\to 0^+} \frac{f'(1/t)(-1/t^2)}{g'(1/t)(-1/t^2)} \qquad \text{(by L'Hôpital's Rule for finite } a\text{)} \\
&= \lim_{t\to 0^+} \frac{f'(1/t)}{g'(1/t)} \\
&= \lim_{x\to\infty} \frac{f'(x)}{g'(x)}
\end{aligned}$$

8. Suppose now that $\lim_{x\to a} f(x)$ and $\lim_{x\to a} g(x)$ are both ∞ (or both $-\infty$).

(a) Suppose $\lim_{x\to a} \frac{f(x)}{g(x)} = L$, where $L \neq 0, \pm\infty$. Write $L = \lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{1/g(x)}{1/f(x)}$ and conclude that $L = \lim_{x\to a} \frac{f'(x)}{g'(x)}$.

Since both $\lim_{x\to a} f(x)$ and $\lim_{x\to a} g(x)$ are both ∞, $\lim_{x\to a} \frac{1}{f(x)}$ and $\lim_{x\to a} \frac{1}{g(x)}$ are both 0, so we have

$$\begin{aligned}
L &= \lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{1/g(x)}{1/f(x)} \\
&= \lim_{x\to a} \frac{-1/(g(x))^2 g'(x)}{-1/(f(x))^2 f'(x)} \qquad \text{(by L'Hôpital's Rule for indeterminate forms of type } 0/0\text{)} \\
&= \lim_{x\to a} \frac{(f(x))^2}{(g(x))^2} \frac{g'(x)}{f'(x)} \\
&= \lim_{x\to a} \frac{(f(x))^2}{(g(x))^2} \lim_{x\to a} \frac{g'(x)}{f'(x)} \\
&= \lim_{x\to a} \left(\frac{f(x)}{g(x)}\right)^2 \lim_{x\to a} \frac{g'(x)}{f'(x)} \\
&= \left(\lim_{x\to a} \frac{f(x)}{g(x)}\right)^2 \lim_{x\to a} \frac{g'(x)}{f'(x)} \\
&= L^2 \lim_{x\to a} \frac{g'(x)}{f'(x)}
\end{aligned}$$

which implies $L(1-L\lim_{x\to a}\frac{g'(x)}{f'(x)})=0$. Since $L\neq 0$, this implies $1=L\lim_{x\to a}\frac{g'(x)}{f'(x)}$. This further implies $\lim_{x\to a}\frac{g'(x)}{f'(x)}\neq 0$, and therefore

$$\begin{aligned} L &= \frac{1}{\lim_{x\to a}\frac{g'(x)}{f'(x)}} \\ &= \frac{\lim_{x\to a}1}{\lim_{x\to a}\frac{g'(x)}{f'(x)}} \\ &= \lim_{x\to a}\frac{1}{\frac{g'(x)}{f'(x)}} \\ &= \lim_{x\to a}\frac{f'(x)}{g'(x)}. \end{aligned}$$

(b) Now suppose $\lim_{x\to a}\frac{f(x)}{g(x)}=L=0$. Apply the result in part (a) to the function $\frac{f(x)+g(x)}{g(x)}$ to conclude that $L=\lim_{x\to a}\frac{f'(x)}{g'(x)}$.
Since both $\lim_{x\to a}f(x)=\infty$ and $\lim_{x\to a}g(x)=\infty$, $\lim_{x\to a}(f(x)+g(x))=\infty+\infty=\infty$. Note also that

$$\lim_{x\to a}\frac{f(x)+g(x)}{g(x)}=\lim_{x\to a}\frac{f(x)}{g(x)}+\lim_{x\to a}\frac{g(x)}{g(x)}=0+1=1\neq 0.$$

By part (a),

$$\begin{aligned} \lim_{x\to a}\frac{f(x)+g(x)}{g(x)} &= \lim_{x\to a}\frac{f'(x)+g'(x)}{g'(x)} \\ 1 &= \lim_{x\to a}\frac{f'(x)}{g'(x)}+1 \end{aligned}$$

and therefore $\lim_{x\to a}\frac{f'(x)}{g'(x)}=0=\lim_{x\to a}\frac{f(x)}{g(x)}$.

(c) Finally, suppose $\lim_{x\to a}\frac{f(x)}{g(x)}=L$, where $L=\infty$ (or $-\infty$). Consider $\lim_{x\to a}\frac{g(x)}{f(x)}$ to conclude that $\lim_{x\to a}\frac{f(x)}{g(x)}=\lim_{x\to a}\frac{f'(x)}{g'(x)}$. (*Hint*: Use Cauchy's Mean Value Theorem to determine the sign of $\lim_{x\to a}\frac{f'(x)}{g'(x)}$.)
Suppose $\lim_{x\to a}\frac{f(x)}{g(x)}=\infty$. Then $\lim_{x\to a}\frac{g(x)}{f(x)}=0$. By part (b), $\lim_{x\to a}\frac{g(x)}{f(x)}=\lim_{x\to a}\frac{g'(x)}{f'(x)}=0$. It follows that $\lim_{x\to a}\frac{f'(x)}{g'(x)}=\pm\infty$. (♣ Need to use Cauchy's Mean Value Theorem to show that $\frac{f'(x)}{g'(x)}$ has the same sign as $\frac{f(x)}{g(x)}$. Also, neeto to add case where $x=\pm\infty$.)

□